

Heat Transfer

MEP 212



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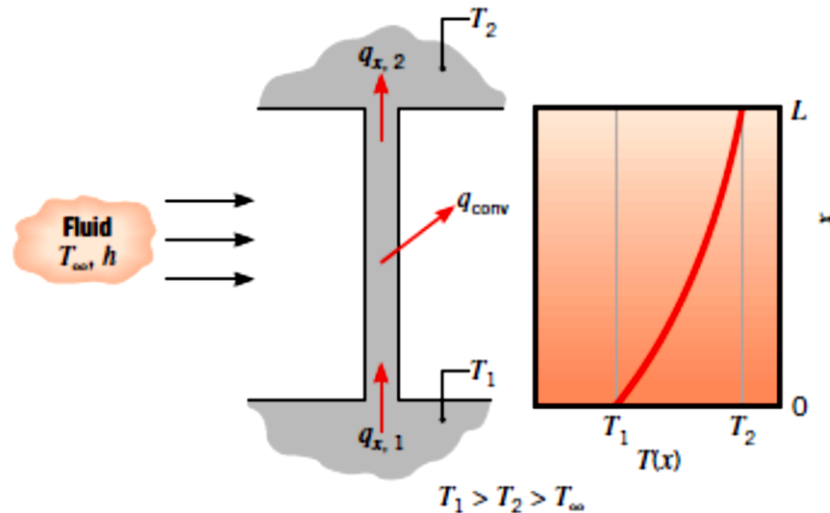


Extended surfaces “Fins”



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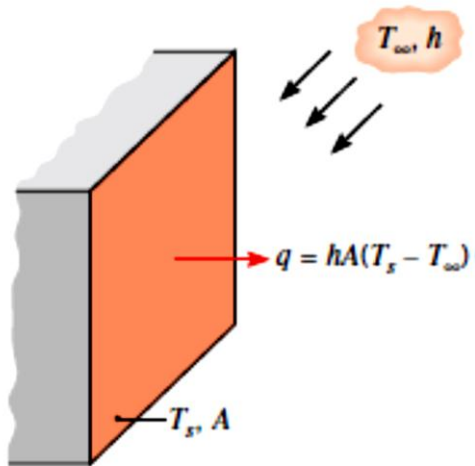
The term ***extended surface*** is commonly used to depict an important special case involving heat transfer by conduction within a solid and heat transfer by convection (and/or radiation) from the boundaries of the solid.



Extended surface “fins”

The extended surface is used to **enhance** the heat transfer rate between a **solid** surface and **an adjoining fluid**.

Consider a plane wall



$$q_{conv} = hA(T_s - T_\infty)$$

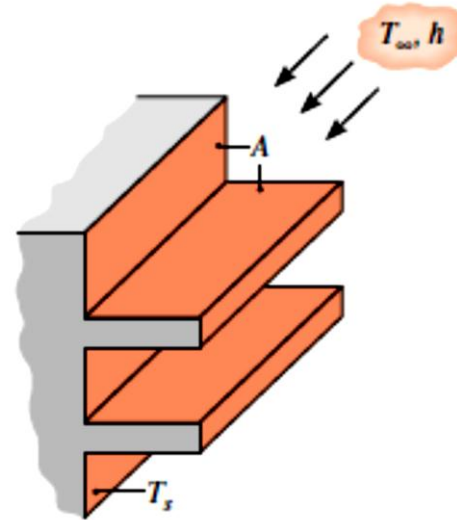
For fixed T_s , two ways to enhance heat transfer rate:

- Increase the fluid velocity. Cost of blower, fan or pump
- Decrease T_∞ : impractical

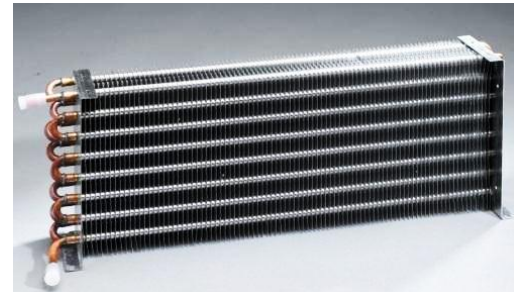
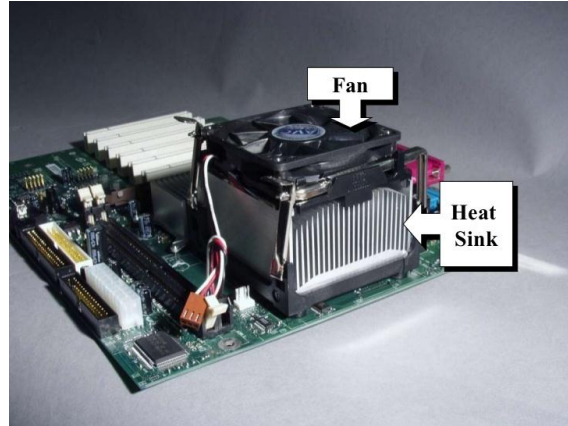
Extended surface “fins”

How about increasing surface area for convection?

By providing fins that extend from the wall into the surrounding fluid.



Fins applications



Fin material

- k of the fin material has a **strong effect** on the temperature distribution along the fin and therefore influences the degree to which the heat transfer rate is **enhanced**.
- Ideally, the fin material should have a **large k** to minimize temperature variations from its base to its tip.
- In the limit of infinite thermal conductivity, the entire fin would be at the temperature of the base surface, thereby providing the maximum possible heat transfer enhancement.

Fin configurations

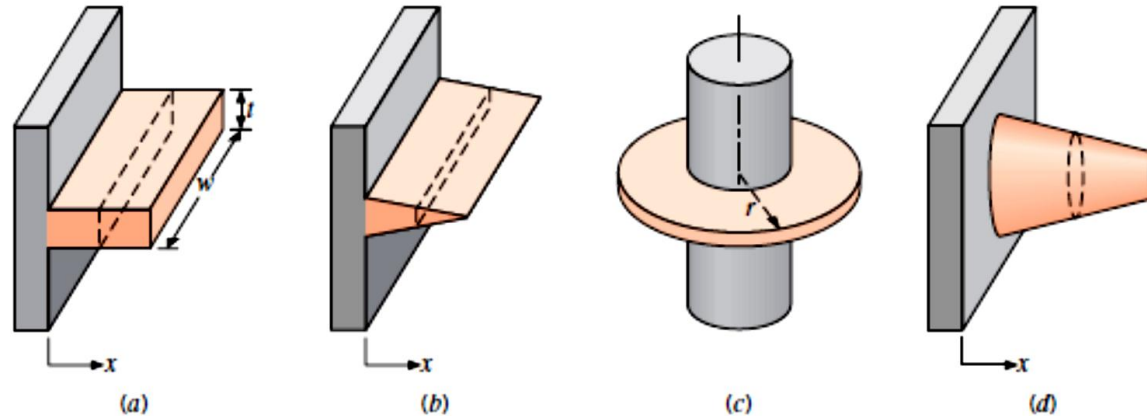
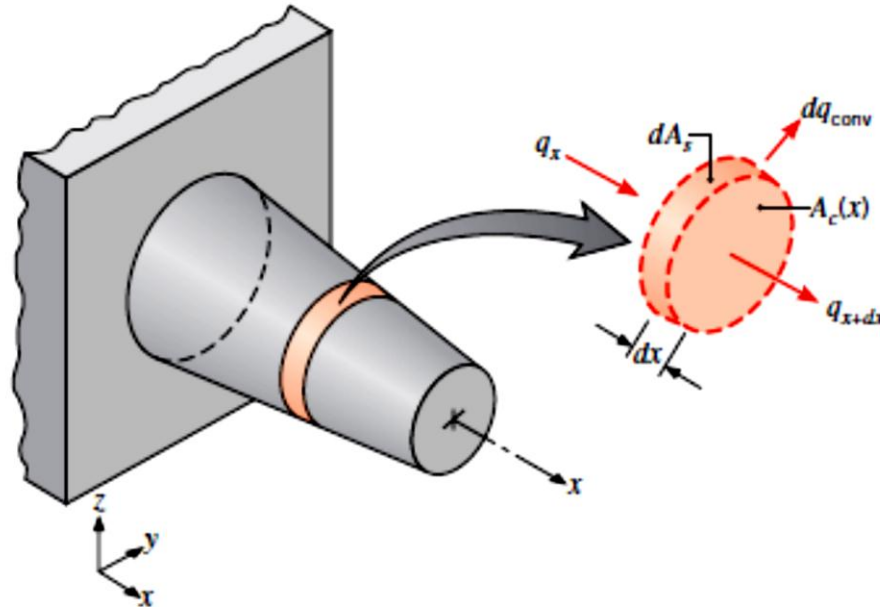


FIGURE 3.15 Fin configurations. (a) Straight fin of uniform cross section. (b) Straight fin of nonuniform cross section. (c) Annular fin. (d) Pin fin.

For an extended surface, the direction of heat transfer from the boundaries is ***perpendicular*** to the principal direction of heat transfer in the solid.

General conduction analysis

To determine the heat transfer rate associated with a fin, we must first obtain the temperature distribution along the fin.



Assumptions

- 1-D heat transfer (longitudinal x direction). In practice the fin is thin and the temperature changes in the longitudinal direction are much larger than those in the transverse direction.
- Steady state
- k is constant
- No heat generation
- Negligible radiation from the surface
- The rate at which the energy is convected to the fluid from any point on the fin surface must be balanced by the rate at which the energy reaches that point due to conduction in the transverse (y, z) direction.
- h is uniform over the surface

Fin analysis

$$q_x = q_{x+dx} + dq_{\text{conv}}$$

$$q_x = -kA_c \frac{dT}{dx}$$

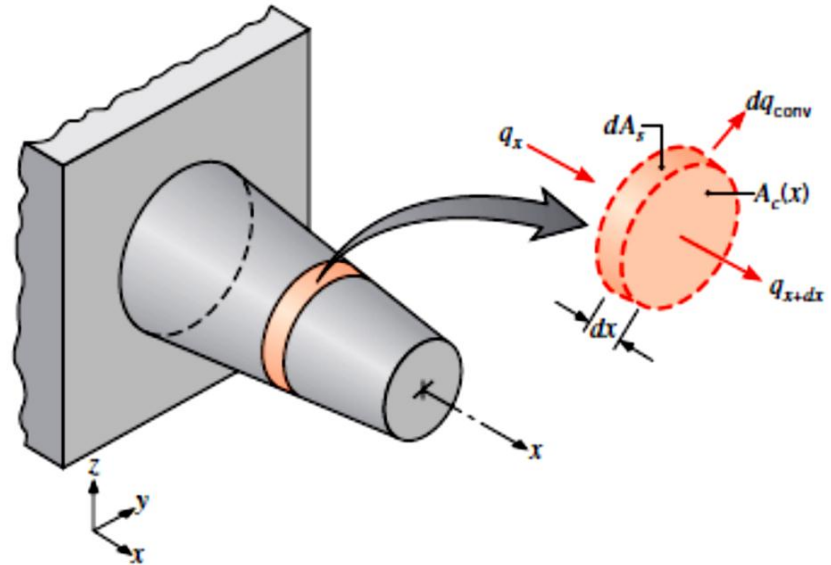
$$q_{x+dx} = q_x + \frac{dq_x}{dx} dx$$

$$q_{x+dx} = -kA_c \frac{dT}{dx} - k \frac{d}{dx} \left(A_c \frac{dT}{dx} \right) dx$$

$$dq_{\text{conv}} = h dA_s (T - T_\infty)$$

By substitution

$$\frac{d}{dx} \left(A_c \frac{dT}{dx} \right) - \frac{h}{k} \frac{dA_s}{dx} (T - T_\infty) = 0$$



$$\frac{d^2T}{dx^2} + \left(\frac{1}{A_c} \frac{dA_c}{dx} \right) \frac{dT}{dx} - \left(\frac{1}{A_c} \frac{h}{k} \frac{dA_s}{dx} \right) (T - T_\infty) = 0$$



Fin of uniform cross-section

$$\frac{d^2 T}{dx^2} - \frac{hP}{kA_c} (T - T_\infty) = 0$$

$$\theta(x) \equiv T(x) - T_\infty$$

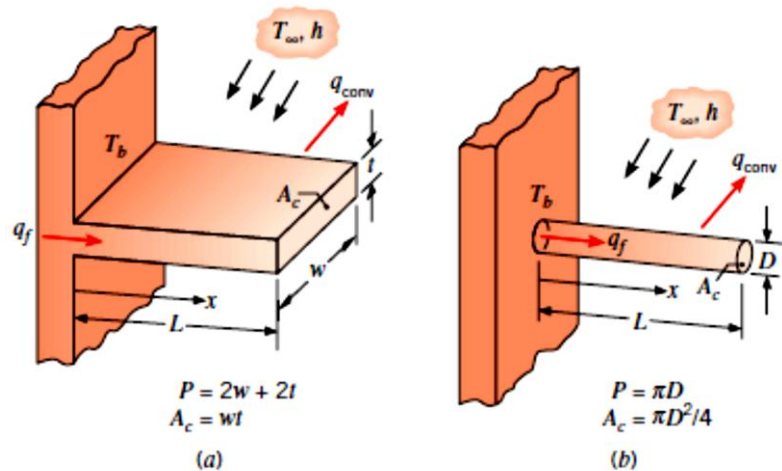
$$m^2 \equiv \frac{hP}{kA_c}$$

$$\frac{d^2 \theta}{dx^2} - m^2 \theta = 0$$

The general solution of this second order differential equation is

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$

It is necessary to specify appropriate BCs for C_1 and C_2 .



Boundary conditions:

One such condition may be specified in terms of the temperature at the *base* of the fin ($x = 0$)

$$\theta(0) = T_b - T_\infty \equiv \theta_b$$

The second condition, specified at the fin tip ($x = L$), may correspond to any one of the four different physical conditions:

- A. Convection heat transfer from the fin tip
- B. Adiabatic condition at the fin tip
- C. Prescribed temperature maintained at the fin tip
- D. Infinite fin (very long fin)

Boundary conditions:

A. Infinite fin (very long fin):

$$\text{As } L \rightarrow \infty, \theta_L \rightarrow 0$$

B. Adiabatic condition at the fin tip:

$$\left. \frac{d\theta}{dx} \right|_{x=L} = 0$$

C. Convection heat transfer at the fin tip:

$$hA_c[T(L) - T_\infty] = -kA_c \left. \frac{dT}{dx} \right|_{x=L} \implies h\theta(L) = -k \left. \frac{d\theta}{dx} \right|_{x=L}$$

D. Prescribed temperature maintained at the fin tip:

$$\theta(L) = \theta_L$$

Case (A): Infinite fin (very long fin)

As $L \rightarrow \infty$, $\theta_L \rightarrow 0$ and $e^{-mL} \rightarrow 0$

$$\begin{aligned}\theta|_{x=0} &= \theta_b; & \theta|_{x=L} &= 0 \\ \theta_b &= C_1 e^{mx} + C_2 e^{-mx}; & C_1 e^{mL} + C_2 e^{-mL} &= 0\end{aligned}$$

$$\frac{\theta}{\theta_b} = e^{-mx}$$

$$q_f = -kA_c \left. \frac{d\theta}{dx} \right|_{x=0} = \sqrt{hPkA_c} \theta_b$$

Case (B): Adiabatic tip fin “negligible heat transfer from tip”

$$\theta|_{x=0} = \theta_b; \quad \left. \frac{d\theta}{dx} \right|_{x=L} = 0$$

$$\theta_b = C_1 e^{mx} + C_2 e^{-mx}; \quad C_1 e^{mx} - C_2 e^{-mx} = 0$$

$$\frac{\theta}{\theta_b} = \frac{e^{mx}}{1 + e^{2mL}} + \frac{e^{-mx}}{1 + e^{-2mL}}$$

$$\frac{\theta}{\theta_b} = \frac{\cosh[m(L - x)]}{\cosh mL}$$

Case (B): Adiabatic tip fin “negligible heat transfer from tip”

$$\begin{aligned} q_f &= -kA_c \left. \frac{dT}{dx} \right|_{x=0} = -kA_c \left. \frac{d\theta}{dx} \right|_{x=0} \\ &= -kA_c \quad m\theta_b \left[\frac{1}{1 + e^{2mL}} - \frac{1}{1 + e^{-2mL}} \right] \\ &= \sqrt{hPkA_c} \theta_b \left[\frac{e^{mL} - e^{-mL}}{e^{mL} + e^{-mL}} \right] \end{aligned}$$

$$q_f = \sqrt{hPkA_c} \theta_b \tanh mL$$

Case (C): Convection From fin tip

$$\theta|_{x=0} = \theta_b;$$

$$\theta_b = C_1 e^{mx} + C_2 e^{-mx};$$

$$hA_c[T(L)-T_\infty] = -kAc \left. \frac{dT}{dx} \right|_{x=L} \implies h\theta(L) = -k \left. \frac{d\theta}{dx} \right|_{x=L}$$

$$h\theta(L) = -k \left. \frac{d\theta}{dx} \right|_{x=L}$$

Case (C): Convection From fin tip

$$\theta_b = C_1 + C_2$$

$$h(C_1 e^{mL} + C_2 e^{-mL}) = km(C_2 e^{-mL} - C_1 e^{mL})$$

$$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x) + (h/mk) \sinh m(L-x)}{\cosh mL + (h/mk) \sinh mL}$$

$$q_f = q_b = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = -kA_c \left. \frac{d\theta}{dx} \right|_{x=0}$$

$$q_f = \sqrt{hPkA_c} \theta_b \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$$

Summary of equations

TABLE 3.4 Temperature distribution and heat loss for fins of uniform cross section

Case	Tip Condition ($x = L$)	Temperature Distribution θ/θ_b	Fin Heat Transfer Rate q
A	Convection heat transfer: $h\theta(L) = -k d\theta/dx _{x=L}$	$\frac{\cosh m(L-x) + (h/mk) \sinh m(L-x)}{\cosh mL + (h/mk) \sinh mL} \quad (3.75)$	$M \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL} \quad (3.77)$
B	Adiabatic: $d\theta/dx _{x=L} = 0$	$\frac{\cosh m(L-x)}{\cosh mL} \quad (3.80)$	$M \tanh mL \quad (3.81)$
C	Prescribed temperature: $\theta(L) = \theta_L$	$\frac{(\theta_L/\theta_b) \sinh mx + \sinh m(L-x)}{\sinh mL} \quad (3.82)$	$M \frac{(\cosh mL - \theta_L/\theta_b)}{\sinh mL} \quad (3.83)$
D	Infinite fin ($L \rightarrow \infty$): $\theta(L) = 0$	$e^{-mx} \quad (3.84)$	$M \quad (3.85)$
$\theta \equiv T - T_\infty$ $\theta_b = \theta(0) = T_b - T_\infty$ $m^2 \equiv hP/kA_c$ $M \equiv \sqrt{hPkA_c} \theta_b$			



Thank you

